ZZ-T-351C
March 4, 1976
SUPERSEDING
Int. Fed. Spec. ZZ-T-00351b(GSA-FSS)
August 6, 1962 and
Fed. Spec. ZZ-T-351
January 8, 1935

FEDERAL SPECIFICATION

TIP, CANE AND CRUTCH, RUBBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a rubber tip for use on canes or crutches.

1.2 Classification. The rubber tip covered by this specification shall be furnished in the sizes indicated in table I (see 3.4), as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 601 - Rubber: Sampling and Testing.

FSC 6515

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Tip. The tip shall be natural rubber, synthetic rubber, or a mixture of the two. Natural rubber shall include reclaimed natural rubber.

3.1.2 Disk. The disk molded in the rubber tip shall be made of metal.

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3.2 Design and construction. The tip shall be molded in one piece with a disk molded between the bearing surface for the cane or crutch shaft and the bottom wearing surface to resist cutting and penetration of the cane or crutch through the rubber. The outside walls shall be either smooth or corrugated. The inside surface shall be corrugated or formed to grip the cane or crutch shaft securely.

3.2.1 Bottom surface. The bottom wearing surface shall have concentric suction rings forming concave surfaces of varying levels for positive traction.

3.3 Performance.

3.3.1 Resistance to abrasion. The tip shall have an abrasive index of not less than 35 when tested as specified in 4.5.1.

3.3.2 Accelerated aging. The tip shall be non-marking and shall not show brittleness or tackiness when tested as specified in 4.5.2.

3.3.3 Hardness. The hardness of the rubber compound shall be 55 plus or minus 5 when tested as specified in 4.5.3.

3.4 Dimensions. Dimensions of the various sizes of tips shall be in accordance with table I.

TABLE I. Tip sizes and dimensions

Size designation	Inside diameter (inches)	Overall height - minimum (inches)	Inside 1/ depth - maximum (inches)	Diameter of base - minimum (inches)
15	3/8 + 0 - 1/32	1	--	--
16	1/2 + 0 - 1/32	1-1/8	--	--
17	5/8 + 0 - 1/32	1-1/4	--	--
18a	3/4 + 0 - 1/16	1-5/8	--	1-3/8

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TABLE I. Tip sizes and dimensions (cont'd)

Size designation	Inside diameter (inches)	Overall height - minimum (inches)	Inside 1/ depth - maximum (inches)	Diameter of base - minimum (inches)
18b	3/4 + 0 - 1/16	2-7/8	2-1/4	2
19a	7/8 + 0 - 1/16	1-3/4	--	--
19b	7/8 + 0 - 1/16	2-7/8	--	--
20	1 + 0 - 1/16	1-3/4	--	--
21	1-1/8 + 0 - 1/16	1-3/4	--	--

1/ Measured from top of the tip to the deepest innermost surface of the rubber.

3.5 Color. The color of the tip shall be tan, maroon, natural or other light color. The color shall be as specified (see 6.2).

3.6 Identification marking. The tip shall be permanently and legibly marked with the name or registered trademark of the contractor.

3.7 Workmanship. Workmanship shall be first class throughout. The tip shall be free from all defects that detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herien. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the rubber may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Lot. A lot shall consist of all of the tips of the same size, produced from the same material under the same conditions and offered for acceptance at the same time.

4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table II. The unit of product for examination purposes shall be one tip.

TABLE II. Sampling for examination

Type defect	Inspection level	AQL (percent defective)
Major	S-4	1.0
Minor	S-4	2.5
Dimensional	S-3	2.5

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4.3.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table III. The unit of product for test purposes shall be one tip. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE III. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective
Resistance to abrasion	3.3.1	4.5.1	S-4 1/	2/
Accelerated aging	3.3.2	4.5.2	S-4 1/	2/
Hardness	3.3.3	4.5.3	S-4 1/	2/

1/ The sample size shall not be less than 4.
2/ Acceptance number shall be zero for all sample sizes.

4.4 Examination. The tip shall be examined to determine compliance with requirements of this specification.

4.4.1 Dimensional. Examination shall be made to determine compliance with the applicable dimensional requirements. Any dimension that is not within the specified tolerance shall be classified as a defect.

4.4.2 For defects. Examination shall be conducted in accordance with table IV. Examination shall not be restricted to the possible defects listed in table IV.

TABLE IV. Classification of defects

Classification	Defect
Major	
101	Not correct material.
102	Does not contain disk.
103	Disk not molded in the rubber.
104	Inside surface not corrugated or formed to grip cane or crutch shaft securely.
105	Bottom wearing surface does not have concentric suction rings.

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TABLE IV. Classification of defects (cont'd)

Classification	Defect
Minor	
201	Material not a light color.
202	Not free of molding flash.
203	Identification marking not permanent or legible.

4.4.3 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 expressed in units of percent defective.

TABLE V. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.4.4 Examination of closure and reinforcement for containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure and reinforcement shall be in accordance with the appendix to that specification.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.5.1 Resistance to abrasion. The tip shall be tested in accordance with method 14111 of Fed. Test Method Std. No. 601 in order to determine compliance with 3.3.1.

4.5.2 Accelerated aging. The tip shall be exposed to oxygen in accordance with method 7111 of Fed. Test Method Std. No. 601 except the length of exposure to oxygen shall be 48 ± 1/2 hours instead of 94 ± 1/2 hours. At the end of the exposure period, the tip shall be removed from the aging chamber and allowed to condition for 24 hours at room temperature. At the end of the conditioning period the tip shall be examined for tackiness and brittleness. The edge of the wearing surface of the tip shall be drawn across a sheet of white paper to determine compliance with 3.3.2. If only a slight mark, which is easily rubbed off with the fingers, is made on the paper, the tip shall be considered as non-marking.

4.5.3 Hardness. The hardness of the rubber tip shall be as specified in 3.3.3 when tested in accordance with method 3021 of Fed. Test Method Std. No. 601 using a Shore durometer (scale A).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Twelve cane or crutch tips, which constitutes a unit of issue of one package, shall be packaged in a snug-fitting box constructed in accordance with PPP-B-566, PPP-B-636, class domestic, or PPP-B-676. Closure of box shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Level A. Twenty four unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twenty four unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall conform to method II as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings specified in the contract or order the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tip covered by this specification is intended for use on canes or crutches.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. National stock number.
c. Size (see 1.2 and 3.4).
d. Color (see 3.5).
e. Applicable levels of packaging, packing and marking. (see 5.1, 5.2 and 5.3).

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6.3 This specification includes the following items:

National stock number	Item identification
6515-01-013-3912	TIP, CANE AND CRUTCH, 12s - 1-5/8 in. high, 3/4 in. inside dia., 1-3/8 in. base dia., min.
6515-01-013-3911	TIP, CANE AND CRUTCH, 12s - 2-7/8 in. high, 3/4 in. inside dia., 2 in. base dia., min.

6.4 Supersession data. NSN 6515-01-013-3911 replaces NSN 6515-00-777-7355 and NSN 6515-01-013-3912 replaces NSN 6515-00-774-0010.

MILITARY CUSTODIANS:
Army - MD
Navy - MS
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Preparing Activity:
DSA-DM

Civil Agency Coordinating Activities:
GSA-FSS
HEW-HSM
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DOCUMENT IDENTIFIER AND TITLE
ZZ-T-351C TIP, CANE AND CRUTCH, RUBBER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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